

TECHNICAL REPORT

Special Section: PFAS in Agroecosystems—Sources, Impacts, and Opportunities for Mitigating Risks to Human and Ecological Health

Simulating PFAS transport in effluent-irrigated farmland using PRZM5, LEACHM, and HYDRUS-1D models

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Abstract

Application of wastewater effluent to agricultural lands can serve as a sustainable approach to meet irrigation and nutrient needs for crop production. While nitrogen and phosphorous loadings can be effectively managed, concerns have been raised regarding the fate of emerging contaminants, including per- and polyfluoroalkyl substances (PFAS), which are widely detected in wastewater effluent. The objective of this paper was to evaluate the ability of three unsaturated flow and transport models, Pesticide Root Zone Model 5 (PRZM5), LEACHM, and HYDRUS-1D, to predict the distribution of PFAS in the soil profile at the Pennsylvania State University Living Filter site, which has received daily wastewater effluent applications for several decades. The models were modified to include adsorption at the air–water interface (AWI), which has been shown to be an important factor governing PFAS transport and phase distribution in the vadose zone. Simulations showed that PRZM5 did not accurately reproduce the observed perfluorooctanesulfonic acid (PFOS) behavior, which was attributed to the “tipping bucket” approach used for water flow that results in the disappearance of AWI during water flow. In contrast, both LEACHM and HYDRUS-1D captured the observed retention of PFOS and perfluorooctanoic acid (PFOA) over a 50-year simulation period. Due to differences in the approach used to calculate the AWI area, LEACHM predicted greater accumulation of PFOS and PFOA at the AWI compared to HYDRUS-1D. These findings indicate that mathematical models that directly account for unsaturated water flow and adsorption at the AWI are able to provide reasonable predictions of long-term PFAS leaching resulting from land application of wastewater effluent.

Abbreviations: AWI, air–water interface or interfacial; HFPO-DA, hexafluoropropylene oxide dimer acid; OC, organic carbon; PFAS, per- and polyfluoroalkyl substances; PFBA, perfluorobutanoic acid; PFHxS, perfluorohexane sulfonate; PFNA, perfluorononanoic acid; PFOA, perfluorooctanoic acid; PFOS, perfluorooctanesulfonic acid; TDS, total dissolved solids.

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Plain Language Summary

The use of treated wastewater for irrigation and as a nutrient source is widely practiced in sustainable agricultural systems, particularly in arid regions. However, there is the potential for persistent contaminants present in treated wastewater, including the “forever chemicals” per- and polyfluoroalkyl substances (PFAS), to be released into the environment. The goal of this work was to evaluate the ability of three mathematical models to predict the transport of PFAS in soils that had been irrigated with treated wastewater effluent. Two of the models, LEACHM and HYDRUS-1D, were able to capture the measured soil concentrations of PFAS, while one of the models greatly overpredicted migration through the subsurface. These findings demonstrate the importance of correctly describing unsaturated water flow and accounting for contaminant adsorption at the air–water interface.

1 | INTRODUCTION

Agriculture use of freshwater sources (e.g., rivers and aquifers) accounts for approximately 70% of the water demand in arid and semi-arid regions, and is one of the major contributors to global water scarcity (Rout & Kattumuri, 2022; Ventura et al., 2019). To address agricultural water requirements in a sustainable manner, treated wastewater is widely applied to crop lands for irrigation and as a supplemental nutrient source (Farhadkhani et al., 2018; Mishra et al., 2023). Despite the benefits associated with the reuse of treated wastewater, concerns have been raised related to the potential for contamination of crops due to the uptake and accumulation of heavy metals, as well as exposure to pathogens (Ofori et al., 2021). Additionally, there are concerns related to the uptake and accumulation of persistent organic contaminants, and in particular, per- and polyfluoroalkyl substances (PFAS). For example, Huff et al. (2020) investigated the uptake of six PFAS from irrigation water by eight herbaceous and seven woody plant species. Their observations showed that accumulation of PFAS mass in above ground tissue ranged from 3.8% for perfluorooctanesulfonic acid (PFOS) in *Festuca rubra* (red fescue) to as high as 42% for perfluoropentanoic acid in *Schedonorus arundinaceus* (tall fescue). In most cases, short-chain length PFAS, such as perfluorobutanoic acid (PFBA), exhibited greater translocation factors in plants compared to longer chain length homologues (Wang et al., 2020). For example, Lan et al. (2018) reported that PFBA accumulation in roots and shoots of wheat was 20 and 44 times greater than that observed for perfluorooctanoic acid (PFOA).

Due to the adverse health effects associated with PFAS, the U.S. Environmental Protection Agency recently passed maximum contaminant levels (MCLs) in drinking water for PFOA and PFOS at 4.0 parts per trillion (ppt) (US Environmental Protection Agency, 2024). The US-EPA also set MCLs at 10 ppt for perfluorohexane sulfonate (PFHxS), per-

fluorononanoic acid (PFNA), and hexafluoropropylene oxide dimer acid (HFPO-DA), and a hazard index of unity for a mixture of two or more for PFNA, PFHxS, HFPO-DA, and perfluorobutane sulfonate (PFBS). Thus, accurate predictions of PFAS fate and transport in the vadose zone at field sites receiving regular applications of treated wastewater are critical in evaluating potential for groundwater contamination. In addition to solid-phase sorption processes that impact most organic chemicals in the subsurface, PFAS exhibit surface-active behavior due to their hydrophilic and hydrophobic moieties. PFAS' surface-active characteristics are exemplified by documented accumulation at the air–water interface (AWI) in porous media, accumulation that is enhanced with increasing C-F chain length and ionic strength (Costanza et al., 2020, 2019; Fabregat-Palau et al., 2022; Liao et al., 2022). Consequently, in variably saturated soils, accumulation at the AWI can influence PFAS migration and alter soil water drainage due to changes in the surface tension of water (Arshadi et al., 2024; Brusseau et al., 2021; Karagunduz et al., 2015; Silva, Šimůnek, et al., 2022).

Several mathematical models have been used to simulate PFAS fate and transport at the field scale, but most of these studies involve parameter sensitivity analyses or hypothetical scenarios rather than comparisons to actual PFAS field data. Shin et al. (2011) used PRZM (Pesticide Root Zone Model), a pesticide leaching model developed by the US-EPA, to retrospectively simulate the environmental distribution of PFOA released from the Washington Works facility in Parkersburg, West Virginia. However, no attempt was made to validate the model or to account for PFAS accumulation at the AWI. Guo et al. (2020) developed a one-dimensional model that incorporated adsorption at the AWI to simulate the transport of PFOS in sand and soil under two hypothetical climate regimes (semiarid and humid) with periodic releases of PFOS. Their model predicted substantial (>70%) retention of PFOS at the AWI, which resulted in minimal migration of PFOS.

Silva et al. (2022; 2020) modified an existing unsaturated flow and transport model, HYDRUS (Šimůnek et al., 2016), to include effects of adsorption at the AWI and corresponding changes in soil water retention. The model was validated using laboratory-scale column data and then used to evaluate the contributions of solid-phase sorption and adsorption at the AWI. In contrast to the results of Guo et al. (2020), sorption by the solid phase was shown to be the dominant mechanism influencing PFAS transport in soils with high sorption capacities (Silva et al., 2020). Wallis et al. (2022) updated LEACHM, a one-dimensional field-scale model originally developed for pesticide leaching, to predict PFAS fate and transport at a fire training site contaminated with aqueous film forming foam (AFFF). Model simulations indicated that rainfall events enhance downward migration of PFAS since the mass adsorbed on the AWI was at a minimum due to the high soil water content.

As briefly summarized above, previous mathematical modeling applications reveal (a) uncertainty regarding the relative importance of adsorption processes versus water flow on PFAS fate and transport in the vadose zone and (b) the limited extent of field-scale validation of available models. The goal of this research was to evaluate three widely available field-scale leaching models, HYDRUS-1D, PRZM5, and LEACHM, to predict PFAS migration in the unsaturated zone. Measured PFAS soil concentration profiles, obtained from the Pennsylvania State University “Living Filter” field site, where irrigation with treated wastewater has been ongoing since the early 1980s, were used to evaluate the model simulations. When necessary, the mathematical models were modified to incorporate non-linear formulations to describe adsorption to the solid phase and at the AWI.

2 | MATERIALS AND METHODS

2.1 | Study site

The study site, also known as “The Living Filter,” is located in State College, Pennsylvania. This site operates as a wastewater treatment facility for the Penn State University and local residences, repurposing treated wastewater for irrigation and groundwater replenishment. Established in the 1960s, the irrigation process occurs daily, with total applications averaging 157.5 cm/year. Soil property details at this site are available in prior studies (Jahn et al., 2023) and were used as input for the model simulations (Table S1). Due to data gaps in meteorological records within the historic simulation period (1965–2014), meteorological data from 2005 to 2014 were sourced from National Oceanic and Atmospheric Administration datasets (<https://www.ncei.noaa.gov/>) and utilized repeatedly for each 10 years of simulation for both historic and “future” (2015–2034) simulations. Soil profile concentrations of PFOA and PFOS for the two time-series simulations were

Core Ideas

- Perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS) are retained near the soil surface following wastewater effluent irrigation.
- Existing transport models were modified to include adsorption at the air–water interface.
- Two models (LEACHM and HYDRUS-1D) provided reasonable predictions of PFOS and PFOA transport.
- PRZM5, using a simplified “tipping bucket” approach, overpredicted PFOS transport.
- Air–water interfacial and solid-phase sorption are critical for accurate PFAS transport predictions.

obtained from previous studies (Jahn et al., 2023; Mroczko et al., 2022) and are shown in Table S2.

2.2 | Mathematical models

Three mathematical models known for their ability to predict contaminant leaching in the vadose zone were selected for modeling PFAS transport at the study site: PRZM5 (Young & Fry, 2016), LEACHM (Hutson & Wagenet, 1989), and HYDRUS-1D (Šimůnek et al., 1998, 2005; Šimůnek et al., 2016). These models were selected due to their demonstrated abilities to predict contaminant transport in the vadose zone and the availability of the source code. Several additional models were considered; however, the inability to modify the source code eliminated these models from further consideration.

To incorporate PFAS adsorption at the AWI, modifications were made to PRZM5 and HYDRUS-1D source codes and the programs were recompiled. Under equilibrium conditions, solute transport can be described as follows:

$$\theta \frac{\partial C}{\partial t} + \rho_b \frac{\partial S}{\partial t} + \frac{\partial (K_{aw} A_{aw} C)}{\partial t} = \frac{\partial}{\partial z} \left(\theta D_H \frac{\partial C}{\partial z} \right) - q \frac{\partial C}{\partial z} \quad (1)$$

where C is the PFAS aqueous concentration (M/L^3), S is the PFAS solid phase concentration (M/M), K_{aw} is the air–water interfacial sorption coefficient (L^3/L^2), A_{aw} is the air–water interfacial area (L^2/L^3), θ is the volumetric water content ($-$), ρ_b is the soil bulk density (M/L^3), D_H is the hydrodynamic dispersion coefficient (L^2/t), and q is the Darcy velocity (L/t).

The AWI adsorption coefficient (K_{aw}) was computed using (Guo et al., 2020):

$$K_{aw} = \frac{\sigma_0 b}{RT (C + a)} \quad (2)$$

TABLE 1 Key input parameters used in the model simulations.

Time	Irrigation rate (cm/year)	Wastewater effluent concentration (ng/L)		AWI sorption parameter ^a , AWI sorption parameter ^a , b (–)				Solid phase sorption ^b , log (K_{oc}) (cm ³ /g)	
		PFOS	PFOA	PFOS	PFOA	PFOS	PFOA	PFOS	PFOA
50 years (1965–2014)	157.5	75	1000	3.4e-3	1.16e-2	0.107	0.033	3.27	2.29
20 years (2015–2034)	157.5	5.1	7	3.4e-3	1.16e-2	0.107	0.033		
	157.5	5.1	N.A. ^c	3.4e-3	N.A. ^c	0.107	N.A. ^c		N.A. ^c
	78.8	5.1	N.A. ^c	2e-3	N.A. ^c	0.094	N.A. ^c		

Abbreviations: AWI, air–water interface; PFOA, perfluorooctanoic acid; PFOS, perfluorooctanesulfonic acid.

^aAWI sorption parameters were obtained from Costanza et al. (2019) and Guo et al. (2020).

^bSorption parameters were obtained from Jahn et al. (2023).

^cN.A., not applicable. The last two rows show the input parameters for the sensitivity analysis on PFOS transport.

where R is the universal gas constant (J/K/mol), T is the temperature (K), a (mol/m³) and b (–) are the fitting parameters in the Szyszkowski equation (Costanza et al., 2019). Experimental data for PFOA and PFOS adsorption at the AWI were obtained from surface tension measurements reported by Costanza et al. (2019). The area of the AWI was calculated using the following equation (Costanza-Robinson et al., 2008):

$$A_{aw} = 6 \times \frac{(1 - \theta_s)}{d_{50}} \times \left(0.9031 - 0.9112 \times \frac{\theta}{\theta_s} \right) \quad (3)$$

These modifications enabled both models to account for PFAS accumulation at the AWI during transport through the soil profile. The modified LEACHM program, which includes AWI sorption, was obtained from Prof. John Huston. A detailed description of the modifications made to the original version of LEACHM can be found in Wallis et al. (2022). The Living Filter study site simulations were initially conducted over a span of 50 years (1965–2014), utilizing historical PFOA and PFOS effluent concentrations estimated by Jahn et al. (2023), followed by an additional simulation spanning 20 years (2015–2034 projection) using the final state of the 50-year simulation as the starting point and incorporating PFOS and PFOA effluent concentrations measured in 2019–2020 (Mroczko et al., 2022; Table S3). To further assess the impact of parameter changes on PFAS transport and retention behaviors in irrigated soils, a sensitivity analysis was conducted using two additional sets of simulations for PFOS with LEACHM and HYDRUS models. These simulations were also performed for 20 years (2015–2034), using the final “snapshot” of the 50-year historical simulation as the initial conditions, with either a higher AWI sorption coefficient (K_{aw}) or reduced irrigation rates.

Due to the relatively low PFAS concentrations, solid phase sorption was assumed to be linear, and plant uptake of PFOS and PFOA was assumed to be negligible since longer chain length PFAS typically exhibit limited plant uptake and translocation (Huff et al., 2020). Key input parameters used

in the model simulations are shown in Table 1. Additional input parameters, including hydraulic and solid phase sorption parameters, can be found in Table S4.

3 | RESULTS AND DISCUSSION

3.1 | PRZM5 simulations of PFOS leaching

PRZM has long been utilized for modeling pesticide transport in agricultural settings (e.g., Carsel et al., 1985; Pennell et al., 1990). In its original form, PRZM accounted for solute transport considering linear/nonlinear and equilibrium/nonequilibrium adsorption on the solid phase but did not incorporate AWI adsorption. Therefore, the source code was modified to include AWI adsorption and then employed to simulate PFOS transport under conditions without and with AWI adsorption ($K_{aw} = 0$ and $K_{aw} \neq 0$, respectively), as illustrated in Figure 1.

The simulated PFOS aqueous concentration profiles shown in Figure 1a,b indicate that AWI adsorption has a negligible effect on PFAS transport, which contradicts previous studies that highlighted the effects of AWI adsorption on PFAS transport and retention in unsaturated soils (e.g., Guo et al., 2020). The inability of PRZM5 to account for AWI sorption in simulating PFAS transport in the vadose zone can be attributed to the water flow model, which is based on a “tipping bucket” approach, where water only flows to a deeper compartment (soil layer or horizon) once the upper compartment becomes fully saturated with water. However, once the upper compartment reaches complete saturation, the AWI disappears, and thus, there is no longer AWI adsorption.

After 1-year, all compartments in the soil profile became saturated (Figure S1), which was attributed to the combination of irrigation and rainfall (~412 cm/year). Only the near-surface soil horizon exhibited varying water content due to evapotranspiration, and thus, AWI adsorption only occurred in this soil horizon during periods when evapotranspiration was greater than rainfall and irrigation (Figure 1c).

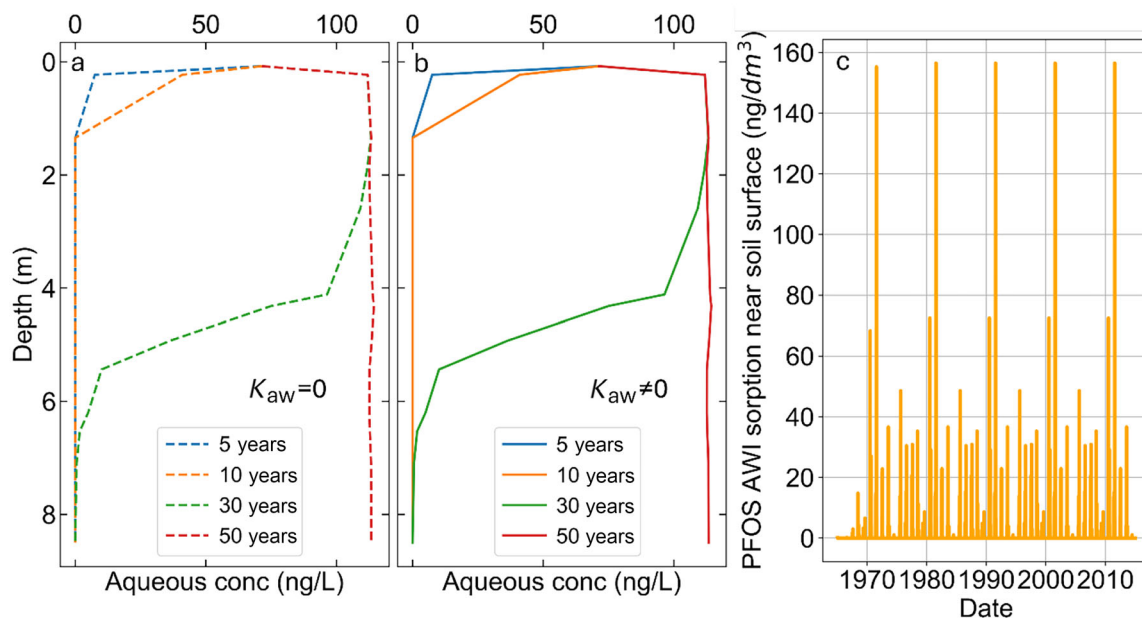


FIGURE 1 Pesticide Root Zone Model 5 (PRZM5) simulation of perfluorooctanesulfonic acid (PFOS) aqueous phase concentrations over a 50-year period (a) without and (b) with air–water interface (AWI) adsorption, and (c) AWI adsorption in near-surface soil horizon (4-cm depth), which occurred when evaporation was greater than precipitation and irrigation.

Consequently, the influence of AWI sorption diminishes as water flows into lower soil horizons, which results in nearly identical PFOS aqueous phase soil profiles (Figure 1a,b). Due to the absence of AWI adsorption resulting from the approach used to describe water flow, no further simulations were performed with PRZM5.

3.2 | LEACHM simulations of PFOA and PFOS leaching

Unlike PRZM5, LEACHM uses Richard's equation to model water flow, which has been shown to provide realistic and reliable simulations of moisture distribution in the vadose zone (e.g., Ng et al., 2000; Pennell et al., 1990). To simulate PFAS transport in variably saturated soils, LEACHM was extended to include AWI sorption as described in Wallis et al. (2022). The updated LEACHM program was used to simulate the transport of PFOA and PFOS at the Penn State Living Filter site. Simulated aqueous phase and solid phase concentrations, without and with AWI adsorption, for a 50-year period are shown in Figure 2. In Figures 2a and 2e, it is apparent that, without considering AWI adsorption, the concentrations of PFOS and PFOA in pore water did not reach the bottom of the simulation zone (10-meter depth) until approximately 30 and 5 years, respectively. Here, PFOS exhibited substantially slower downward migration compared to PFOA, which is consistent with its greater solid-phase adsorption capacity. When the AWI adsorption was accounted for (Figure 2b,f), additional reductions in the mobility of PFOS and PFOA were

observed. Here, PFOS did not reach the bottom of the soil profile over the 50-year simulation, whereas PFOA began to exit at the bottom of the simulated zone after approximately 10 years. The more pronounced effect of AWI adsorption on PFOS migration relative to PFOA is consistent with the greater affinity for PFOS to accumulate at the AWI compared to PFOA (Costanza et al., 2019).

Comparisons of simulated solid-phase PFOS and PFOA concentrations without and with AWI adsorption to those measured at the Living Filter field site (Jahn et al., 2023) are shown in Figure 2c,g,d,h, respectively. LEACHM simulations accurately captured the elevated PFOS and PFOA concentrations detected near the soil surface, which was attributed primarily to a higher organic carbon (OC) content of the shallow soil layers (Table S1). Additionally, the solid phase results show that incorporation of AWI sorption in the simulations yielded minimal differences between observed and predicted PFOS and PFOA solid-phase concentrations over most of the soil profile. However, notable deviations were evident when comparing the measured PFOS concentration at the deepest (~8 m) depth to those simulated without and with AWI adsorption.

3.3 | HYDRUS-1D simulations of PFOA and PFOS leaching

Using identical weather conditions and field site parameters, HYDRUS-1D was used to simulate PFOA and PFOS transport at the Living Filter site. Similar to LEACHM,

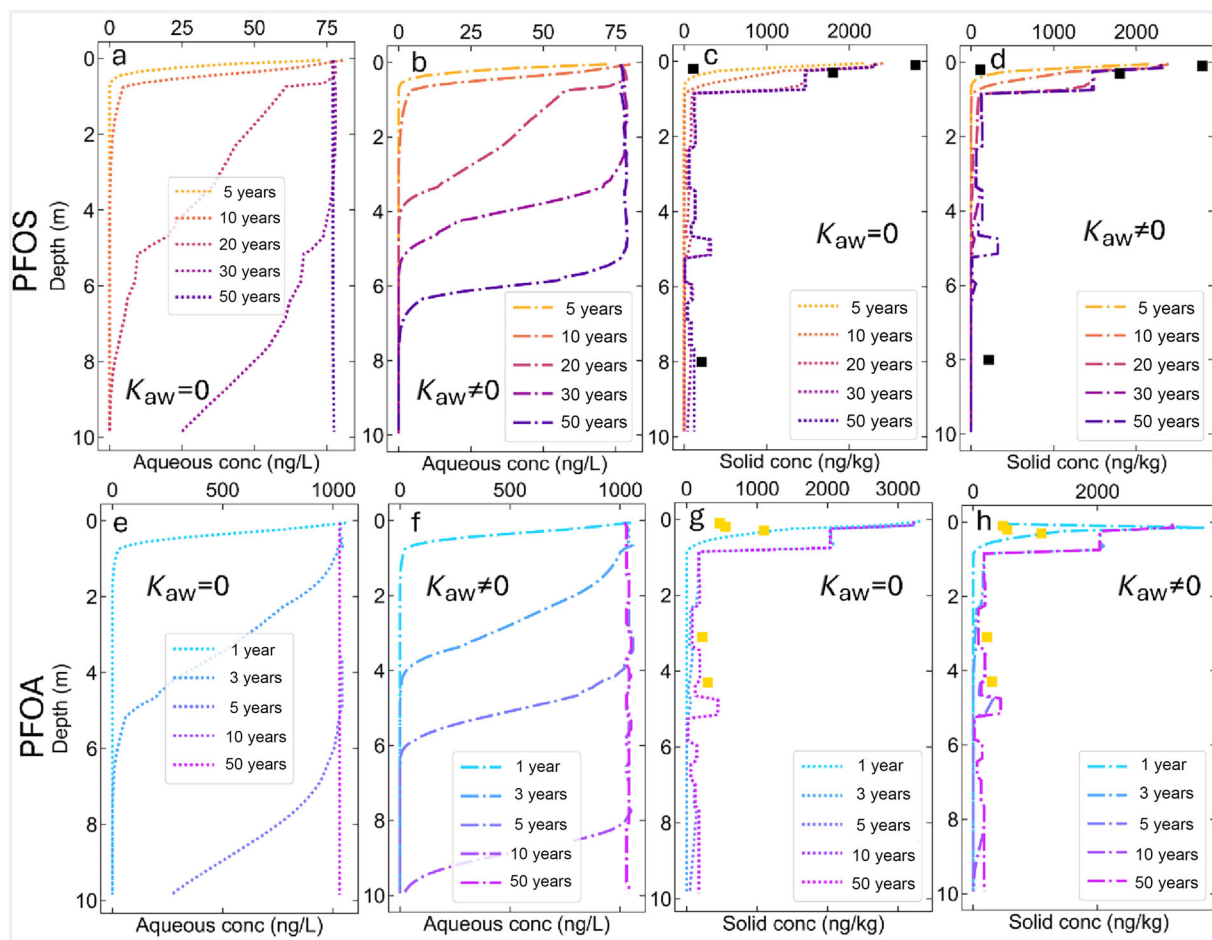


FIGURE 2 Simulations of perfluorooctanesulfonic acid (PFOS) aqueous phase and solid phase profiles over time (a, c) without and (b, d) with air–water interface (AWI) sorption, and perfluorooctanoic acid (PFOA) aqueous phase and solid phase profiles over 50 years (e, g) without and (f, h) with AWI sorption using LEACHM program. The black dots and yellow dots in solid phase profiles represent measured PFOS and PFOA concentrations (Jahn et al., 2023).

HYDRUS-1D employs Richards equation to simulate water flow under variably saturated conditions. To account for adsorption at the AWI, modifications were made to the HYDRUS-1D code (Equations 2 and 3), enabling the calculation of the AWI area and incorporating PFAS accumulation at the AWI. Results of the 50-year simulation are shown in Figure 3 for the aqueous and solid phase concentrations, without and with adsorption at the AWI. Similar to the results obtained with LEACHM, the HYDRUS-1D simulations demonstrate that the inclusion of adsorption at the AWI impedes the downward migrations of PFOS and PFOA (Figure 3a,b,e,f, respectively), resulting in prolonged breakthrough times at a depth of 10 m. A comparison of HYDRUS-1D and LEACHM predictions for PFOS with AWI adsorption shows that this compound reached a depth of 10 m within 40 years (Figure 3b), whereas breakthrough was not observed in the LEACHM simulations even after 50 years (Figure 2b), indicating a significantly higher retardation. However, both models predicted similar PFOS and PFOA solid-phase concentrations profiles (Figure 3c,d,g,h, respec-

tively), thus consistently demonstrating substantial retention near the soil surface. A comparison with the field data indicates that at lower depths, the HYDRUS-1D simulations underpredicted the measured PFOS and PFOA concentrations and exhibited little differences between predictions with and without adsorption at the AWI at later times.

3.4 | Comparison of LEACHM and HYDRUS-1D simulations

To further investigate the effects of AWI adsorption on PFAS transport and retention in unsaturated soils at the study site and to compare the performance of LEACHM and HYDRUS-1D, temporal and spatial distributions of PFOS and PFOA mass across different phases for the two models are shown in Figure 4. Both LEACHM and HYDRUS-1D predicted PFOS accumulation across different phases over a 50-year period (Figure 4a,c,e,g, respectively). Initially (within the first decade), a substantial portion of PFOS was sorbed by the solid

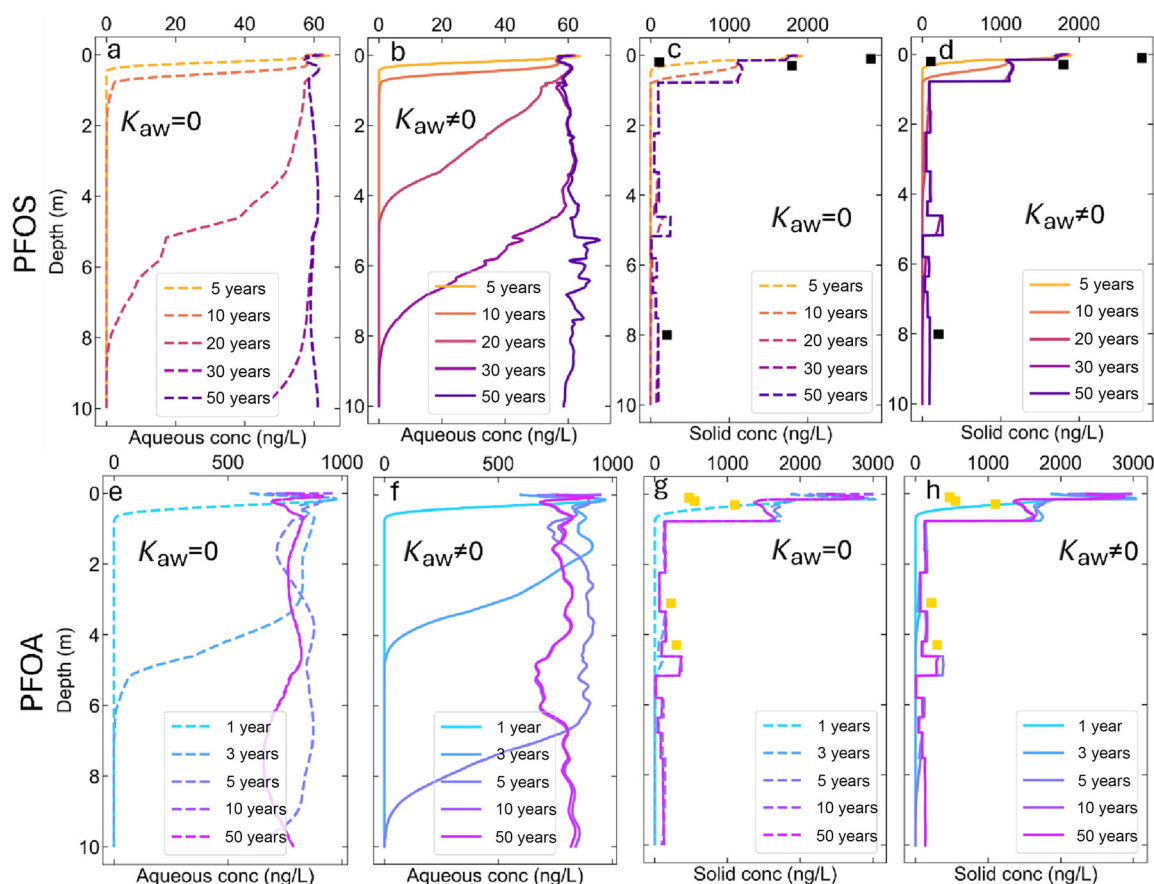


FIGURE 3 Simulation results of perfluorooctanesulfonic acid (PFOS) aqueous phase and solid phase profiles over time (a, c) without and (b, d) with air–water interface (AWI) sorption, and perfluorooctanoic acid (PFOA) aqueous phase and solid phase profiles over 50 years (e, g) without and (f, h) with AWI sorption using HYDRUS-1D. The black and yellow dots in the solid phase profiles represent measured PFOS and PFOA concentrations (Jahn et al., 2023)

phase, primarily concentrated near the soil surface, where OC content is elevated. This behavior is consistent with the time-series of solid-phase distribution shown in Figures 2 and 3. After this initial phase, the fraction of PFOS associated with the solid-phase reached a plateau. Subsequently, the AWI emerged as the primary reservoir for PFOS mass, with PFOS adsorption at the AWI steadily increasing throughout the 50-year simulation period. A snapshot of the final day of the simulation shown in Figure 2b,d reveals that LEACHM predicts AWI sorption occurring primarily in the middle portion of the soil profile, whereas HYDRUS-1D predicts a more uniform distribution of AWI sorption along the depth of the soil profile.

Another notable disparity between LEACHM and HYDRUS-1D simulations lies in the ratio of mass sorbed by the solid phase and mass accumulated at the AWI. LEACHM predicted a larger fraction of PFOS mass at the AWI (~45%) after 50 years, whereas HYDRUS-1D predicted only ~27% of the PFOS mass at the AWI (Figure 4a,c). While differences in some input parameters may have resulted in slightly different predictions of water flow between the two models (see

Figure S1 and Tables S1 and S4), the primary reason for this discrepancy in phase distribution is likely originating from the difference in how the methods compute the AWI area. Specifically, at the same soil water saturation, LEACHM computes a larger AWI area available for PFAS partitioning (Figure S2) and therefore predicts a higher partitioning of PFOS to the AWI compared to HYDRUS-1D. This underscores the importance of accurately determining the AWI area before employing any simulation tool to predict PFAS transport in variably saturated soils. A similar trend was observed for PFOA, where an initial increase in solid-phase mass was followed by increasing adsorption at the AWI, with both solid-phase and AWI sorption approaching an asymptote before the conclusion of the simulation period (Figure 4e,g). LEACHM results also suggest a higher degree of AWI sorption compared to HYDRUS-1D, with the majority of AWI adsorption occurring in deeper soils (Figure 4f vs. h).

As discussed in Jahn et al. (2023), the PFAS concentrations in treated wastewater was substantially lower in recent years due to the phase-out of many PFAS-related products. Therefore, it is of interest to simulate future PFAS transport and

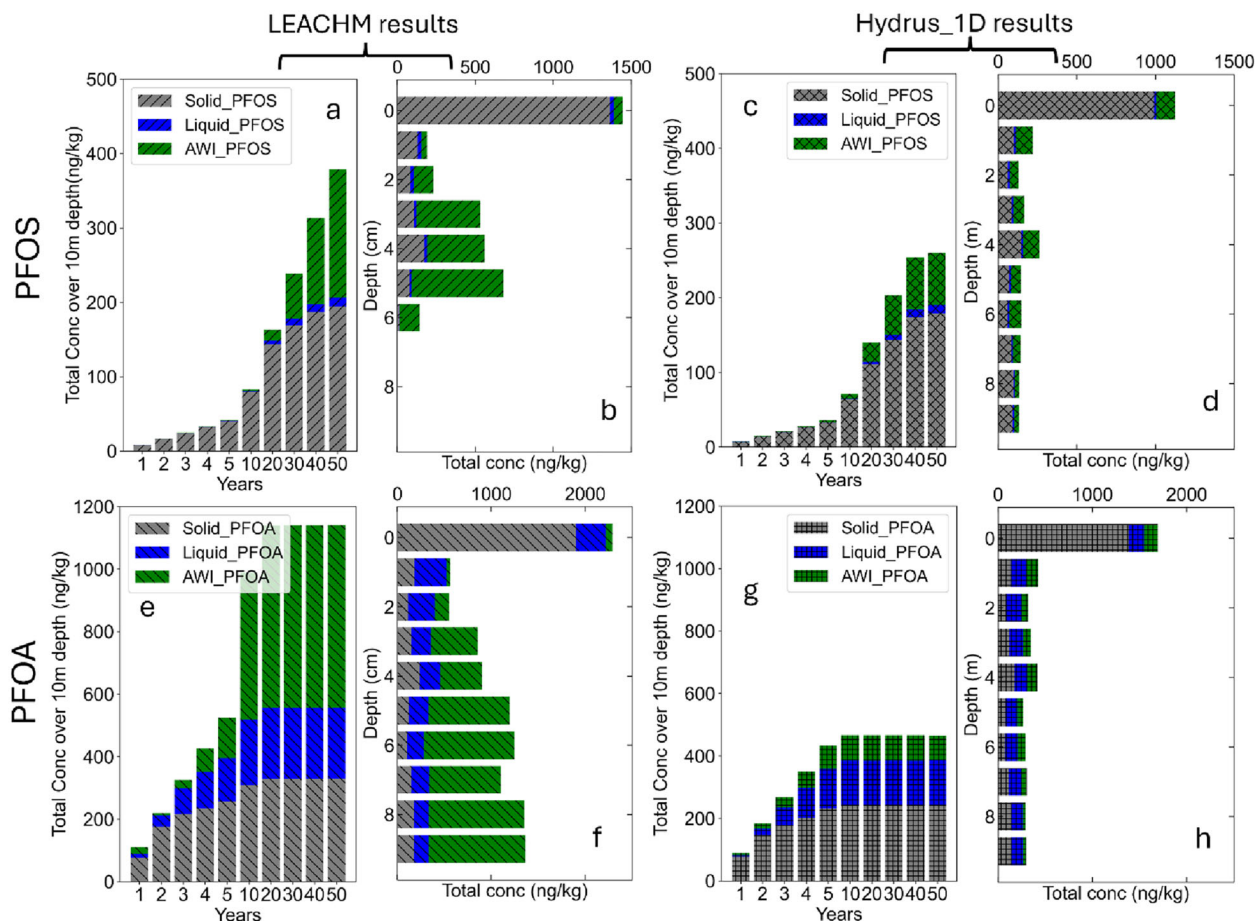


FIGURE 4 LEACHM and HYDRUS-1D predictions of perfluorooctanesulfonic acid (PFOS) (a–d) and perfluorooctanoic acid (PFOA) (e–h) temporal distributions over the 50-year simulation period (a, c, e, g) and spatial distributions at the 50th year (b, d, f, h).

retention profiles at the study site subject to irrigation with the lower PFAS concentration wastewater. Therefore, the simulation was extended for an additional period of 20 years (into the future), utilizing the most recent measurements of PFAS concentration in the wastewater effluent at this site (Table S3).

Both LEACHM and HYDRUS-1D predicted a redistribution of PFOS in both the aqueous and solid phases of the soil profile following the application of a lower PFAS concentration wastewater (Figure 5a,b,d,e, respectively). However, the LEACHM simulation indicated no reduction in total PFOS concentration within the 10-m soil profile (Figure 5d), as the PFOS mass adsorbed to the solid phase was subsequently transferred to the AWI over the 20-year simulation period (Figure 5c,d). In contrast, the HYDRUS-1D simulation predicted a 44% decrease in total PFOS concentration, with most of the loss from the soil profile attributable to solid-phase desorption (Figure 5g,h). These findings indicate that the AWI can play an important role in the retention of PFOS in unsaturated soils. Additionally, PFOS desorption from the solid phase does not necessarily imply downward movement since the released mass can be transferred to the AWI, leading to a prolonged source of PFOS contamination.

3.5 | Model sensitivity analysis

The concentration of total dissolved solids (TDS) in a solution, as well as the irrigation rate, could have significant implications for PFAS adsorption at the AWI. Given that wastewater typically exhibits varying TDS concentrations and that irrigation rates influence soil water content, thus AWI dynamics, it is important for site managers to understand how changes in TDS and irrigation rates may affect PFAS transport and retention in the soil profile. To explore these effects, two 20-year simulations for PFOS were conducted using both LEACHM and HYDRUS with a higher K_{aw} and lower irrigation rates (Figure 6). No discernible differences are observed when comparing simulation results in Figure 6a,b with previous simulations in Figure 5a,d, respectively. Similarly, Figures 6e versus 5e and Figures 6f versus 5h exhibit nearly identical PFOS phase distributions over time. These findings collectively indicate that any changes in TDS (i.e., increased K_{aw}) are likely to have minimal effects on PFOS transport and retention over the long term. However, when considering the irrigation rate, as illustrated in Figures 6c and 6d with LEACHM and Figure 6g,h with HYDRUS, a reduction in

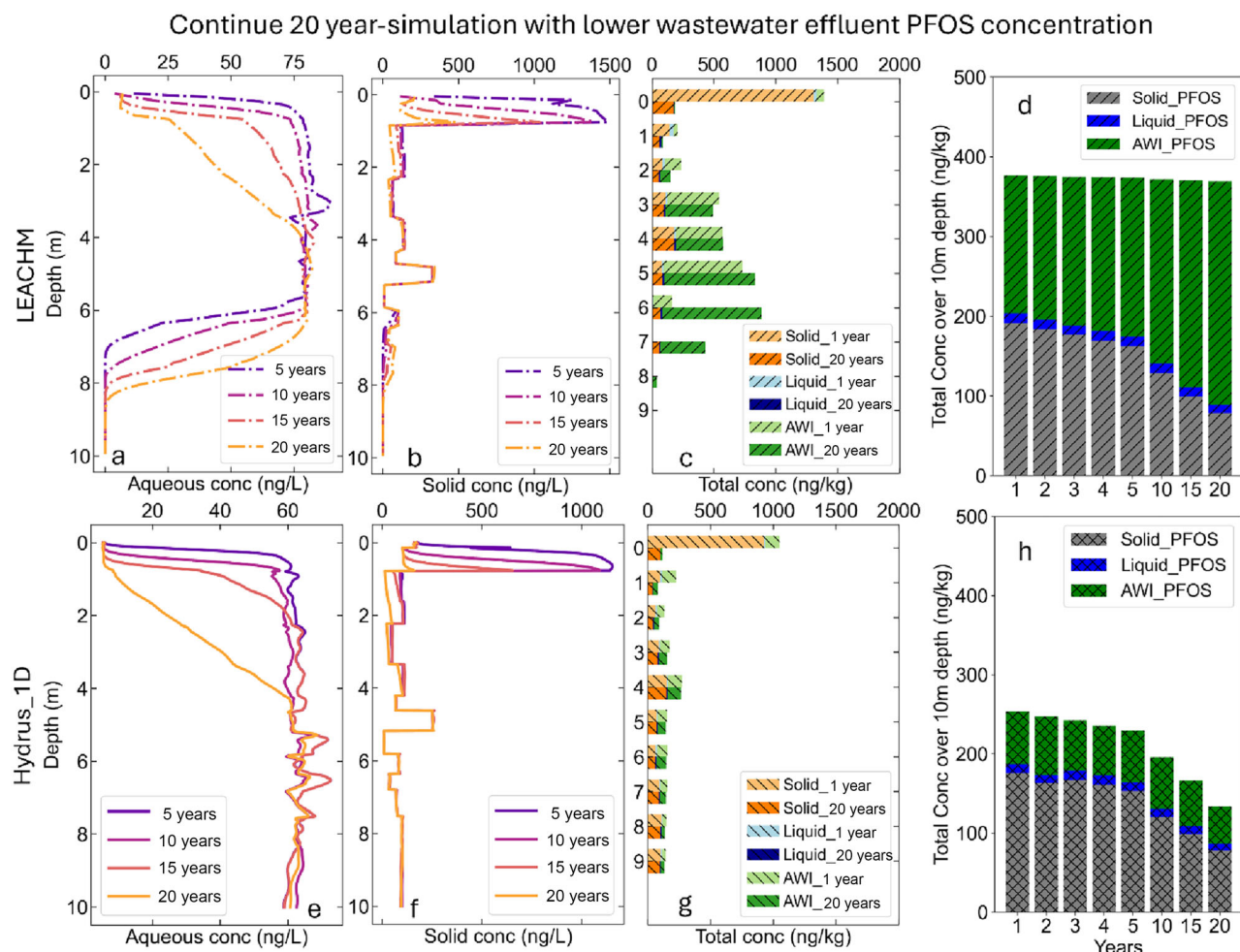


FIGURE 5 LEACHM simulated perfluorooctanesulfonic acid (PFOS) (a) aqueous phase and (b) solid phase profiles over a 20-year future simulation period, (c) PFOS spatial mass distribution in the soil column after 1 year and 20 years, and (d) PFOS temporal mass distribution across phases over the 20-year period. HYDRUS-1D simulated PFOS (e) aqueous phase and (f) solid phase profiles over a 20-year future simulation period, (g) PFOS spatial mass distribution in the soil column after 1 year and 20 years, and (h) PFOS temporal mass distribution across phases over the 20-year period.

irrigation rate (from 157.5 cm/year to 78.4 cm/year) resulted in much slower migration of PFOS through the soil profile. For example, HYDRUS predicted a higher PFOS total concentration in the soil profile after 20 years at a lower wastewater irrigation rate (Figure 6h). These sensitivity analysis results underscore the significance of wastewater irrigation rate over wastewater TDS when predicting potential PFOS migration at the study site.

4 | CONCLUSIONS

This study evaluated the ability of three unsaturated zone leaching models, HYDRUS-1D, LEACHM, PRZM5, to predict PFOA and PFOS leaching at the Penn State Living Filter site. The field site has been irrigated with treated wastewater since the 1980s. Measured PFOS and PFOA concentrations

in the soil profile showed that these PFAS are concentrated in the surface horizons. Due to the tendency for PFOA and PFOS to accumulate at interfaces, the models were modified to include adsorption at the AWI, which was previously shown to influence PFAS transport in the vadose zone. Simulation results showed that PRZM5 was not able to accurately predict PFOS migration, which was a consequence of the “tipping bucket” approach used to describe water flow. This approach requires a soil horizon to be fully saturated with water before the water can move downward in the profile. This saturation condition causes the AWI interface to disappear, resulting in an overestimation of PFAS transport. In contrast, both LEACHM and HYDRUS-1D were able to provide reasonable predictions of the measured PFOS and PFOA soil profile over a 50-year simulation period. However, due to differences in the approach used to calculate the AWI area, LEACHM predicted greater accumulation of PFOS and

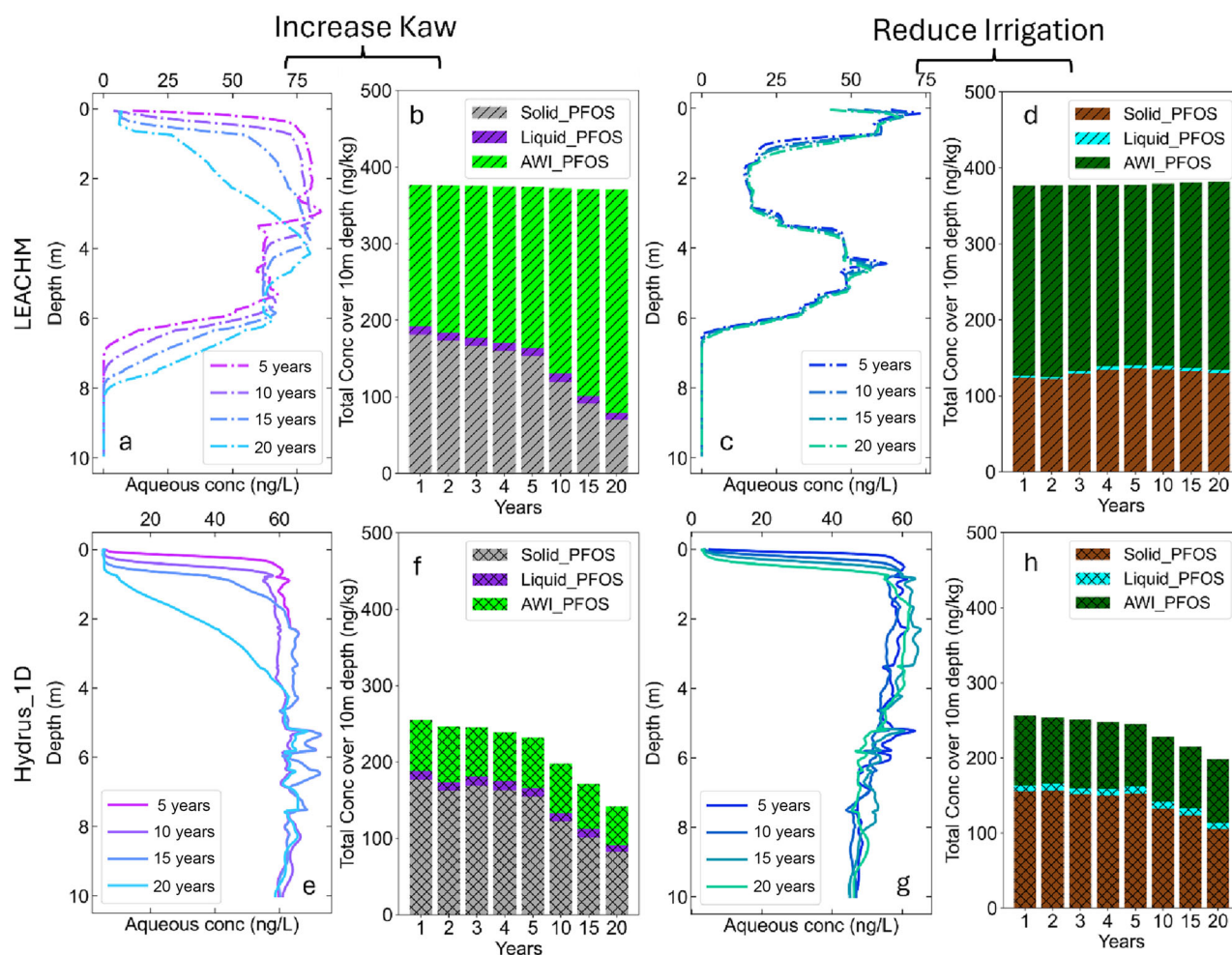


FIGURE 6 LEACHM simulations of perfluorooctanesulfonic acid (PFOS) (a and c) aqueous phase and (b and d) changes in the mass distribution over 20 years with higher K_{aw} and lower irrigation rate, respectively. HYDRUS-1D simulations of PFOS (e and g) aqueous phase and (f and h) mass distribution change over 20 years with higher K_{aw} and lower irrigation rate, respectively.

PFOA in the soil profile compared to HYDRUS-1D. A substantial portion of PFOS and PFOA mass was retained near the soil surface, which was due in large part to the high sorption capacity of the upper soil horizon. Interestingly, the fraction of PFOS and PFOA associated with the solid phase reached a plateau or limit, and subsequently, the adsorption at the AWI steadily increased throughout the 50-year simulation period. These results indicate the importance of sorption processes (solid-phase and AWI) on the accumulation of PFAS in the soil horizon with several decades of treated wastewater irrigation.

AUTHOR CONTRIBUTIONS

Shuchi Liao: Formal analysis; investigation; methodology; software; writing—original draft. **Uriel Garza-Rubalcava:** Writing—reviewing and editing. **Linda M. Abriola:** Supervision; writing—reviewing and editing. **Heather E. Preisen-danz:** Funding acquisition; data curation; writing—reviewing and editing. **Linda S. Lee:** Funding acquisition; writing—

reviewing and editing. **Kurt D. Pennell:** Conceptualization; funding acquisition; methodology; resources; supervision; writing—original draft.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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